# CMAQ Inferential and Predictive Ability Analysis

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**Abstract**

Using the Multivariate Normal Distribution (MVN), I created a covariance matrix structure that describes the spatial correlation within the $O_3$ measurements and CMAQ predictions. I then imputed CMAQ values to create a full data set. A Generalized Least-Squares model was then employed to determine the relationship between CMAQ predictions and $O_3$ values. Predictions at pre-specified sites were then made by imputing $O_3$ measures and CMAQ predictions at the supplied locations, along with prediction intervals to describe our uncertainty. The model provides predictions at the pre-specified sites and describes how $O_3$ levels relate to the know CMAQ values.

## 1 Introduction

Ozone ($O_3$) is a major component of harmful pollutants that infect the air we breath. We seek to advise people of dangerous levels of pollution and identify levels of pollution on a large scale. However, it is not feasible to attempt to measure $O_3$ at every single point in an area.

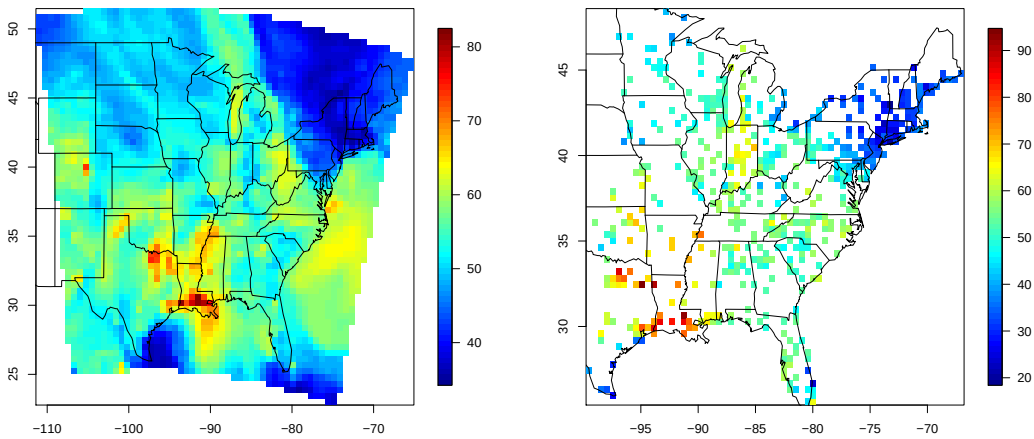
For this reason, the Community Multi-scale Air Quality Model (CMAQ) was created. The model incorporates readily available data to predict $O_3$ levels throughout the Eastern US. The model is known to not work well, but I believe it can be used to predict $O_3$ levels.

For our analysis, I considered data gathered from the Eastern United States. The CMAQ predictions I've been provided are rather dense, as can be seen in Figure 1a. By contrast, the stations we received $O_3$ readings from are sparse, shown in Figure 1b.

Further, consider how the readings match up. Figure 1 shows the $O_3$ readings and the CMAQ predictions from $O_3$ measurement stations and the CMAQ algorithm, respectively. The trends are similar. We hope to show in this analysis how actual $O_3$ readings relate to CMAQ predictions.

The CMAQ model and the $O_3$ data gathering stations are not at the same locations, however. Because of this, we cannot use a simple Generalized Least-Squares model adjusting for correlation. Also, the spatial correlation within the data frustrates our ability to use many common models.

We therefore seek to create a model that can account for the spatial correlation in our data. Using this model, I hope to determine the relationship between the CMAQ model and actual $O_3$ readings. Furthermore, we have a list of sites, illustrated in Figure 2, where we want to make predictions of $O_3$ levels. We hope the model will accurately predict the $O_3$ levels at the provided sites along with a measure of uncertainty.




(a) CMAQ predictions for the Eastern US

(b) $O_3$ measurements for the Eastern US

Figure 1: $O_3$ measurements compared to CMAQ predictions

## 2 Model Description

In considering the data, I thought it reasonable to say that $O_3$ readings near CMAQ predictions are measuring the same atmospheric effects. As such, I assigned all predictions and measurements the same Longitude and Latitude value if CMAQ predictions were within a certain distance from the $O_3$ prediction site. However, not all $O_3$ measurement sites had a CMAQ prediction within the window I specified.

I therefore used the Multivariate Normal Distribution (MVN) to model the data. Such a model is very useful, because it allows us to fill in missing data. In addition, every conditional distribution will follow a normal distribution. In using the MVN, we assume that every factor is distributed normally, and that every relationship between two factors included in the MVN model are linear.

The correlation in our data poses a significant problem. Because every observation is assumed to be correlated with every other, our 800 observed $O_3$ values constitute one draw from the MVN. I defined the MVN in this analysis as follows:

$$\boldsymbol{y} \sim N(\boldsymbol{X}\boldsymbol{\mu}, \Sigma)$$

where $\boldsymbol{y}$ is the $O_3$ measurements and CMAQ predictions from 800 sites. $\boldsymbol{y}$ is created such that each site $i$ has both $O_3$ measurements and CMAQ predictions considered for each site. For $y_{ij}$ in $\boldsymbol{y}$, $i$ denotes the site location, and $j = O_3, CQ$

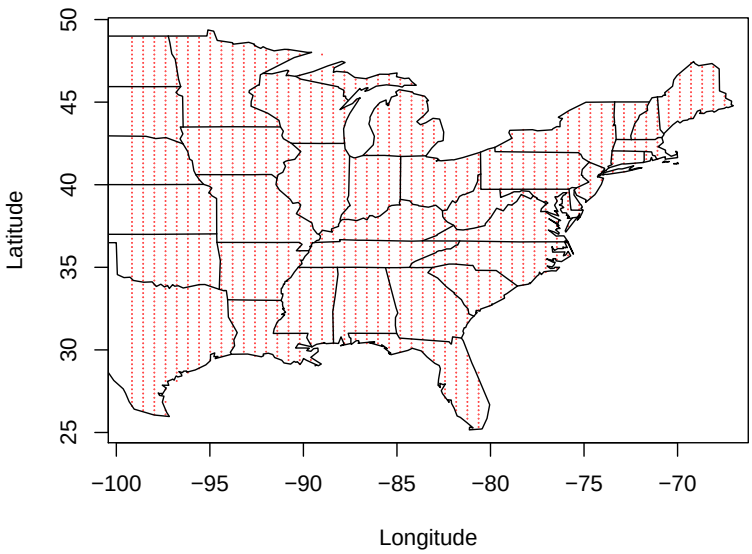



Figure 2: Designated Prediction Sites for the Model

for an ozone reading and a CMAQ prediction, respectively. $\boldsymbol{y}$ would look like

$$\boldsymbol{y} = \begin{bmatrix} y_{1_{O_3}} \\ y_{1_{CQ}} \\ y_{2_{O_3}} \\ . \\ . \\ . \\ y_{800_{CQ}} \end{bmatrix}.$$

Furthermore, $\boldsymbol{\mu}$ is a vector of the mean $O_3$ value and the mean CMAQ prediction, and $\boldsymbol{X}$ is a matrix of two vectors consisting of zeroes and ones. The $\boldsymbol{X\mu}$ factor in our function centers our values around the average value for either the $O_3$ measures or the CMAQ predictions.

$\Sigma$ accounted for the correlation within our data. In $\boldsymbol{y}$, $O_3$ and CMAQ values are paired by location, such that for every two locations at locations $t_1$ and $t_2$, the correlation between them is equal to $\exp\{-\frac{|t_1-t_2|}{\phi}\}$, where $|t_1 - t_2|$ is the distance between $t_1$ and $t_2$.

The correlation between $O_3$ values and CMAQ predictions was also accounted for. $\Delta$ was the correlation matrix between $O_3$ and CMAQ at a single location. We then created $\Sigma$ such that

$$\Sigma = \begin{bmatrix} \Delta & \exp\{-\frac{|t_1-t_2|}{\phi}\}\Delta & \exp\{-\frac{|t_1-t_3|}{\phi}\} & ... \\ \exp\{-\frac{|t_1-t_2}{\phi}\}\Delta & \Delta & \exp\{-\frac{|t_2-t_3|}{\phi}\} & ... \\ \exp\{-\frac{|t_1-t_3|}{\phi}\}\Delta & \exp\{-\frac{|t_2-t_3|}{\phi}\}\Delta & \Delta & ... \\ ... & ... & ... & ... \end{bmatrix}$$

Using the correlation inherent in these data, I filled in the missing CMAQ values in the $O_3$ measurements. Also, I filled in the $O_3$ values for the prediction locations.

Once I had Latitude, Longitude, and CMAQ predictions connected with every $O_3$ reading, I used a Gaussian process to fit a linear model. A Gaussian process assumes all observations in the data follow a multivariate normal distribution. More formally,

$$\boldsymbol{u} \sim N(\boldsymbol{Z\beta}, \sigma^2(1-\omega)\boldsymbol{R} + \omega\mathbf{I})$$

such that

$$u_i = \beta_0 + \beta_1(CMAQ) + \epsilon_i, \epsilon_i \sim (\sigma^2(1-\omega)\boldsymbol{R} + \omega\mathbf{I}) \tag{1}$$

where $\boldsymbol{u}$ is the $O_3$ readings at our measurement stations, and $i$ again refers to the station number.

This resembles multiple linear regression. However, there are significant differences. Notice that $\boldsymbol{Z}$ is different than $\boldsymbol{X}$ above. $\boldsymbol{Z}$ is a matrix with a vector of ones and a vector of CMAQ predictions. The matrix structure allows us to create a model corresponding to Equation 1.

In the distribution of the error term, $\epsilon_i$, we account for the correlation in the data. Every $\epsilon_i$ accounts for the difference between the estimate for $O_3$ and the actual observed value at station $i$. For every two $\epsilon_i$ at locations $m_1$ and $m_2$, the correlation between them is equal to $\boldsymbol{R} = \exp\{-\frac{|m_1-m_2|}{\phi}\}$, where $|m_1 - m_2|$ is the distance between $m_1$ and $m_2$, similar to the correlation scheme in $\Sigma$ above.

Other parameters in the model account for variability. $\sigma^2$ is the variability that is constant over all the data. $\omega$ accounts for the sampling variability inherent in getting $O_3$ readings.

$\beta_1$ in Equation 1 refers to how much the the $O_3$ will change on average, all other factors held constant for every unit increase in a CMAQ prediction. Our analysis attempts to find the true $\beta_1$ because it is the relation between CMAQ predictions and $O_3$ measurements.

The GLS procedure depends on assumptions similar to our assumptions for the MVN model. Specifically, in order to use the GLS model we assume the data are distributed according to a multivariate normal and the variance is constant. Also, because our model is linear, we assume the relation between CMAQ predictions and $O_3$ measurements is linear.

Our model is able to predict $O_3$ values and determine the relation between CMAQ and $O_3$ if we can justify using it on our data.

# 3 Model Justification

The most glaring issues in our data are the spatial correlation between locations and the absence of CMAQ values close to many of our $O_3$ measurements. Because of this, we cannot simply fit a Generalized Least-Squares or linear regression model. Further, the spatial correlation between $O_3$ values complicates the use of a principal components model.

A multivariate normal model is well-justified because the correlation between $O_3$ values can be worked into the analysis. Further, the correlation within the

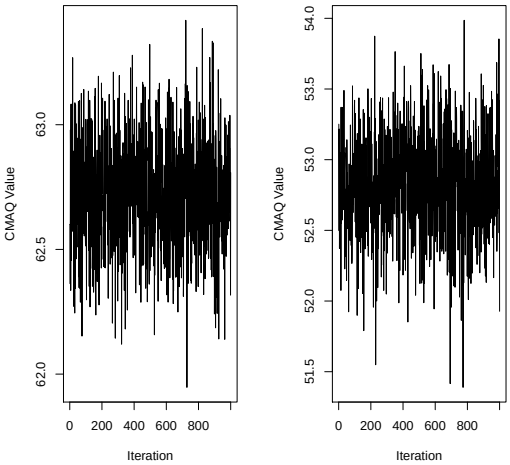



Figure 3: Convergence of Two Sets of Randomly Selected CMAQ Values

multivariate normal distribution allows us to fill in missing data. For these reasons, I considered the multivariate normal more suited to the problems inherent in our data. Using the multivariate normal distribution will also allow us to describe the relationship between CMAQ predictions and $O_3$ values.

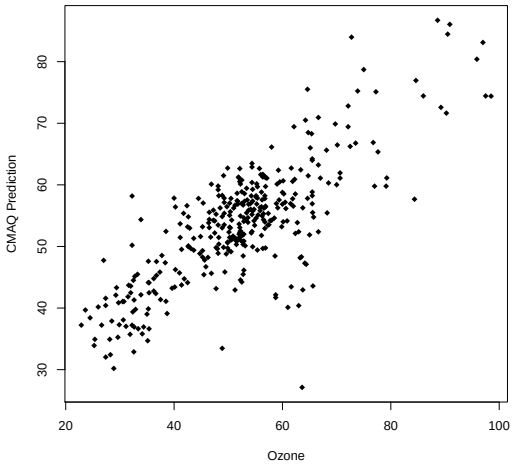
In order assume the data follow a multivariate normal distribution, we have to justify three assumptions: the generated data points converge, the considered variables are normally distributed, and the relation between CMAQ predictions and $O_3$ measurements are linear.

In our data set we had 444 sites without a comparable CMAQ prediction. We generated CMAQ prediction values using the correlation between sites and the generated values will assumedly center around the true value for a CMAQ prediction at that point. Figure 3 shows the behavior of CMAQ predictions that were generated for two sites. It's apparent that the values generally oscillate around a value, such that we can say they converge.

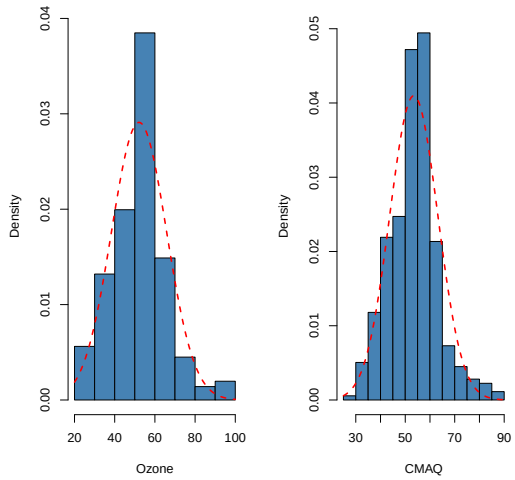
The data are also normal. In Figure 4b we see that $O_3$ measurements and CMAQ predictions are normally distributed. Further, the relationship between CMAQ predictions and $O_3$ readings is linear, as shown in Figure 4a.

A subsidiary assumption to our MVN model deals with how I defined CMAQ predictions and $O_3$ values that were similar enough to be measuring the same atmospheric effects. I assumed that every CMAQ prediction within 0.04 degrees of Longitude and Latitude of our $O_3$ measurement station was similar, such that we considered the CMAQ prediction and $O_3$ measurement to be located at the same place. In doing so, the maximum distance a CMAQ prediction could be from a measurement site was 3.9 miles.

The maximum possible distance between measurement stations and CMAQ prediction locations is not large, but the possible difference in location demands explanation. 44.5% of our $O_3$ readings were matched to similar CMAQ predictions with 0.04 degree distance. This implies we fill in CMAQ prediction values for about half the data set. A smaller window around the measurement stations




(a) Illustration of the Relationship Between CMAQ Predictions and $O_3$ Measurements




(b) Histograms Establishing Normality of the Data

Figure 4

demands that we fill in more CMAQ predictions with less CMAQ predictions and $O_3$ measurement pairs. A smaller window consolidating less than 40% of the data filled in therefore seemed unjustified.

I also assumed a exponential correlation among the data. I did so because the exponential allows for data to be separated by non-integer values and can be based on the distance between two points. All our locations are a certain distance from one another that is not separated into integer distances. Because the distance between locations was not constrained to be a whole number, the exponential correlation structure fit.

Note that $O_3$ levels can only be positive. Usually we would have to constrain a $\boldsymbol{y}$ to be a positive number. However, as our available CMAQ prediction values and $O_3$ values are rather large, such predicted values for locations will not be negative. As can be seen in the Results section, none of our predictions are negative.

The multivariate normal distribution is only part of our model. After we had filled in the missing data, we fit a Generalized Least-Squares model (GLS) with an exponential correlation structure. In order to use this, I assumed the data was normally distributed and had equal variance. The normality of the data was already considered, and the linear relationship between CMAQ predictions and $O_3$ readings was illustrated above. Equal variance was also worked into our multivariate normal model. However, there is no reliable way to justify an assumption of equal variance. Despite this, our assumption that the data is distributed according to a multivariate normal is adequate to justify a GLS model.

Within our GLS model, I included both an Intercept and a CMAQ term due to the highly linear relationship between CMAQ and $O_3$ values. The intercept

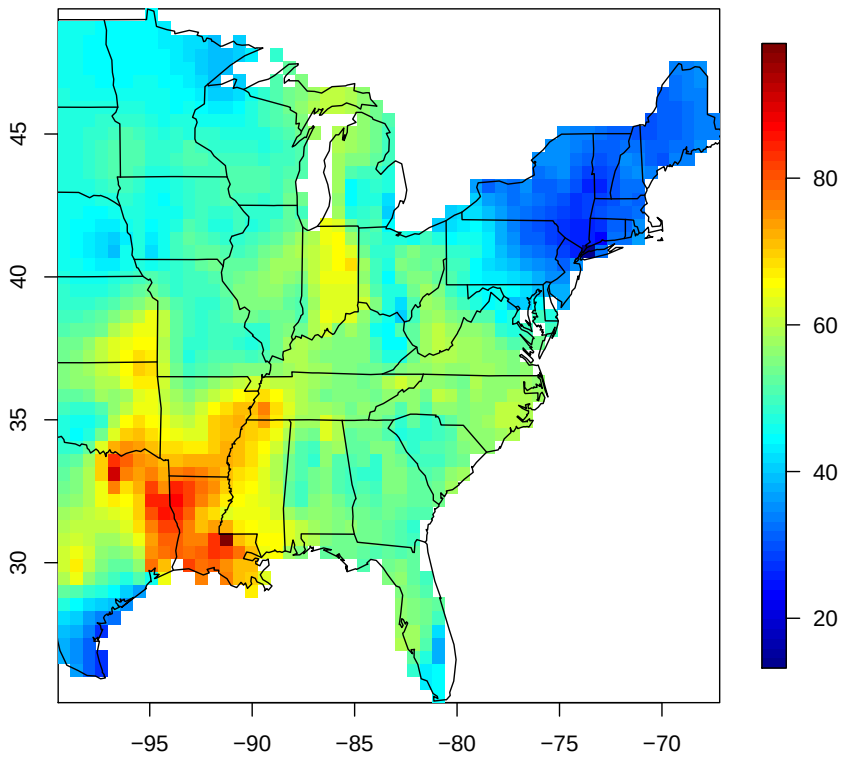



Figure 5: Model Predictions for the Eastern US

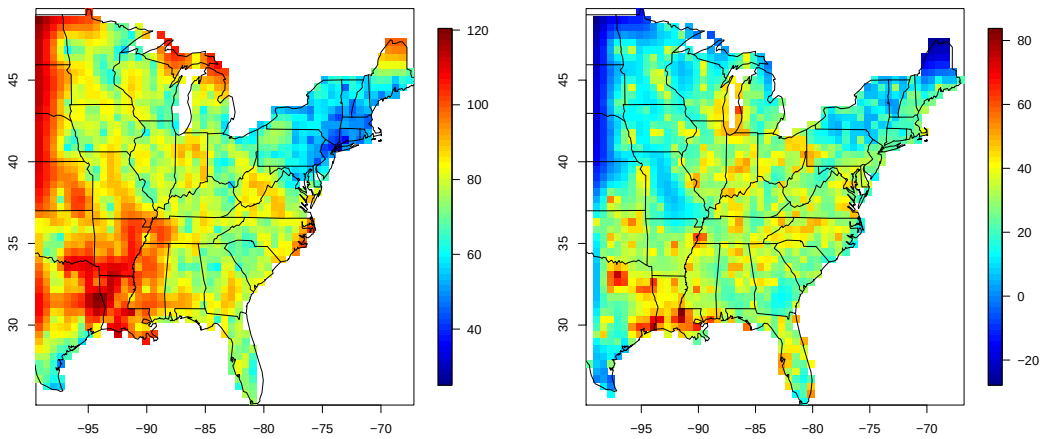
term would then be able to account for the difference in CMAQ and $O_3$ values if our estimate for the effect of CMAQ was close to 1. In order to proceed further with model selection, I needed to compare models at every iteration. Due to the length of time it took to run the MVN model, I was unable to consider another GLS model.

Because of the correlation present in the data, the traditional measure of model fit $R^2$ is of no use to us. Instead, I performed a cross validation study to determine how well the model predicts. Specifically, I removed 10% of the data from my set of $O_3$ measurements and tried to predict the $O_3$ values using my model.

The model predicted quite well, although the estimates of the variance were

| Coverage | RPMSE | Bias |
|---|---|---|
| 0.99 | 5.10 | 0.05 |

Table 1: Cross-Validation Study Results




(a) Upper Prediction Interval Bound for $O_3$ Levels in the Eastern US

(b) Lower Prediction Interval Bound for $O_3$ Levels in the Eastern US

Figure 6

slightly off. Bias evaluates how accurately the model predicts and reports how far off our predictions are on average. Root Predicted Mean Squared Error (RPMSE) reports the average space between the predictions and the actual values. Coverage reports the percentage of observations that were contained in our 95% prediction intervals.

Our cross validation results are listed in Table 1. RPMSE and bias are quite small considering the scale of $O_3$ predictions. We can therefore conclude our model is able to predict accurately. However, the coverage is rather high, such that the variance estimates in our model are too high.

All things considered, our MVN model appears to fit rather well. We are justified in using this model to make further predictions at the sites specified.

# 4 Results

In this model, I sought to determine the relationship between CMAQ predictions and $O_3$ measurements. The GLS model I fit was described above, and the estimates for the $\beta$ coefficients are described in Table 2.

The results are relatively intuitive. As our CMAQ prediction increases by 1, the $O_3$ increases by 1 on average, all other variables held constant. Also, we'd

| Estimate | Lower CI Bound | Upper CI Bound |
|---:|---:|---:|
| -2.54 | -6.10 | 1.02 |
| 1.00 | 0.94 | 1.07 |

Table 2: Estimates and 95% Confidence Intervals for GLS Model Parameters

expect for the $O_3$ value to be -2.54 when our CMAQ prediction is 0.

The confidence interval bounds in Table 2 reflect our uncertainty about the estimates. If the confidence interval is large, we cannot state our estimates with very much certainty. However, if the confidence interval is small we can state that the true value of the relationship between CMAQ and $O_3$ readings is close to $\beta_1$ with a large degree of confidence. For example, we are 95% confident that the true value for the effect of CMAQ on $O_3$ is between 0.94 and 1.07.

We can conclude, then that CMAQ and $O_3$ values should be rather similar and the intercept indicates how biased CMAQ values normally are. Notice that the confidence interval for the Intercept term includes 0, such that we can't determine what the intercept is with a great deal of confidence.

Our predictions for the $O_3$ levels are illustrated in the Figures 5 and 6. Figure 5 shows the model's estimates for $O_3$ levels at the pre-specified locations. As I am not certain of the results due to variability in the data, Figure 6 shows the upper and lower prediction intervals for the model's estimate. Therefore, I am 95% confident that the $O_3$ level at a certain location is between the $O_3$ estimates contained in Figure 6 at that location. The prediction intervals therefore provide a guide of what we can expect $O_3$ values to be at any one of our prediction locations.

# 5 Conclusions

Using a multivariate normal model, I imputed CMAQ prediction values. Using this algorithm, I matched $O_3$ measurements to CMAQ predictions and found the relation between CMAQ predictions and $O_3$ measures using a Generalized Least-Squares model. Also, I used the Multivariate Normal model to make predictions of $O_3$ levels at the pre-specified prediction sites.

Using the techniques described above, I determined the relation between CMAQ predictions and $O_3$ measures, such that CMAQ values are slightly biased estimators of $O_3$ readings. I also created predictions at all the sites supplied, shown in Figures 5 and 6. I also showed in my cross validation study that this model predicts effectively.

The model's prediction of the variance is somewhat off. Also, my model relies on grouping CMAQ predictions and $O_3$ measures which may not be measuring the same effect.

Further steps should be taken to determine what is the smallest window that can be used without sacrificing model accuracy in terms of RPMSE. Such an analysis could take the form of a simulation study. Furthermore, a meteorologist should be consulted to determine how far atmospheric conditions, such as $O_3$ content, can be generalized. More information on these subjects would perhaps provide us with a better model.

I would also suggest a more in-depth inquiry into why the variance estimates are so large. Also, I would suggest performing model selection on our GLS model, such that we can determine whether an intercept term is appropriate to include.